

SkillSight

identifying explicit and implicit skills from resume excerpts.

Multi label classification

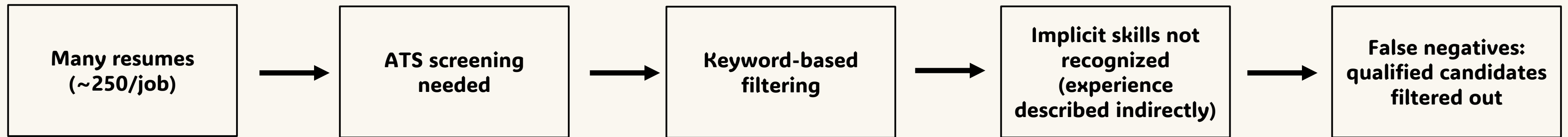
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How resume screening works today?



The system optimizes for keyword matches, not skill fit - so real talent is missed

SkillSight: Making Implicit Skills Visible

- **Implicit tech skills:** detect software/engineering skills from resume/JD context, even if not written.
- **Beyond keywords:** use semantic understanding instead of keyword matching.
- **Clear labels:** separate Explicit vs Implicit technical skills.
- **Synthetic data:** train and evaluate on a curated synthetic dataset of software/engineering resumes and job descriptions.
- **Practical output:** return a focused tech-skill profile to reduce noise and missed candidates.



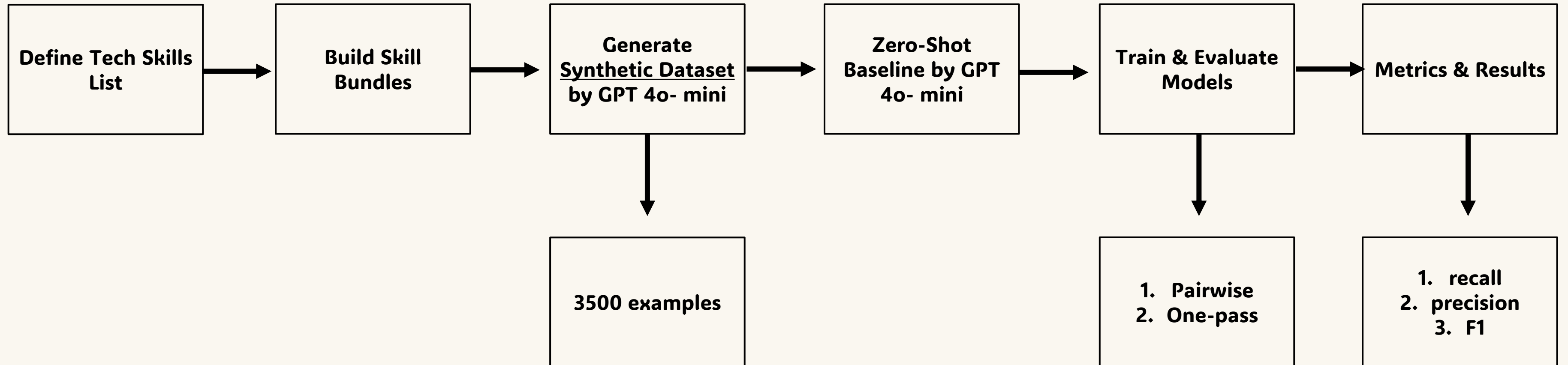
Explicit & Implicit Skills in Tech Resumes: A Quick Example

JD: “I wrote complex **SQL** queries to build a **star schema for analytics**-
defining fact and dimension tables and optimizing joins for reporting.”

Explicit: SQL

Implicit: Dimensional Modeling

SkillSight Pipeline Overview



End-to-End: Data • Models • Metrics

Data:

- 3,500 synthetic, controlled examples (JD-like)
- Fixed global skill list across all samples
- Each example is built around 3–6 skills sampled from the global list (via skill bundles)
- Per-skill labeling: NONE / IMPLICIT / EXPLICIT

Example from the Dataset:

"job_description": "In my previous role.... , ultimately enhancing the overall educational experience we provided.",
"skills": {"Bash": 1.0, "PowerShell": 1.0, "Pulumi": 0.5, "Terraform": 0.5, "Perl": 0.5, "Azure": 0.5}

End-to-End: Data • Models • Metrics

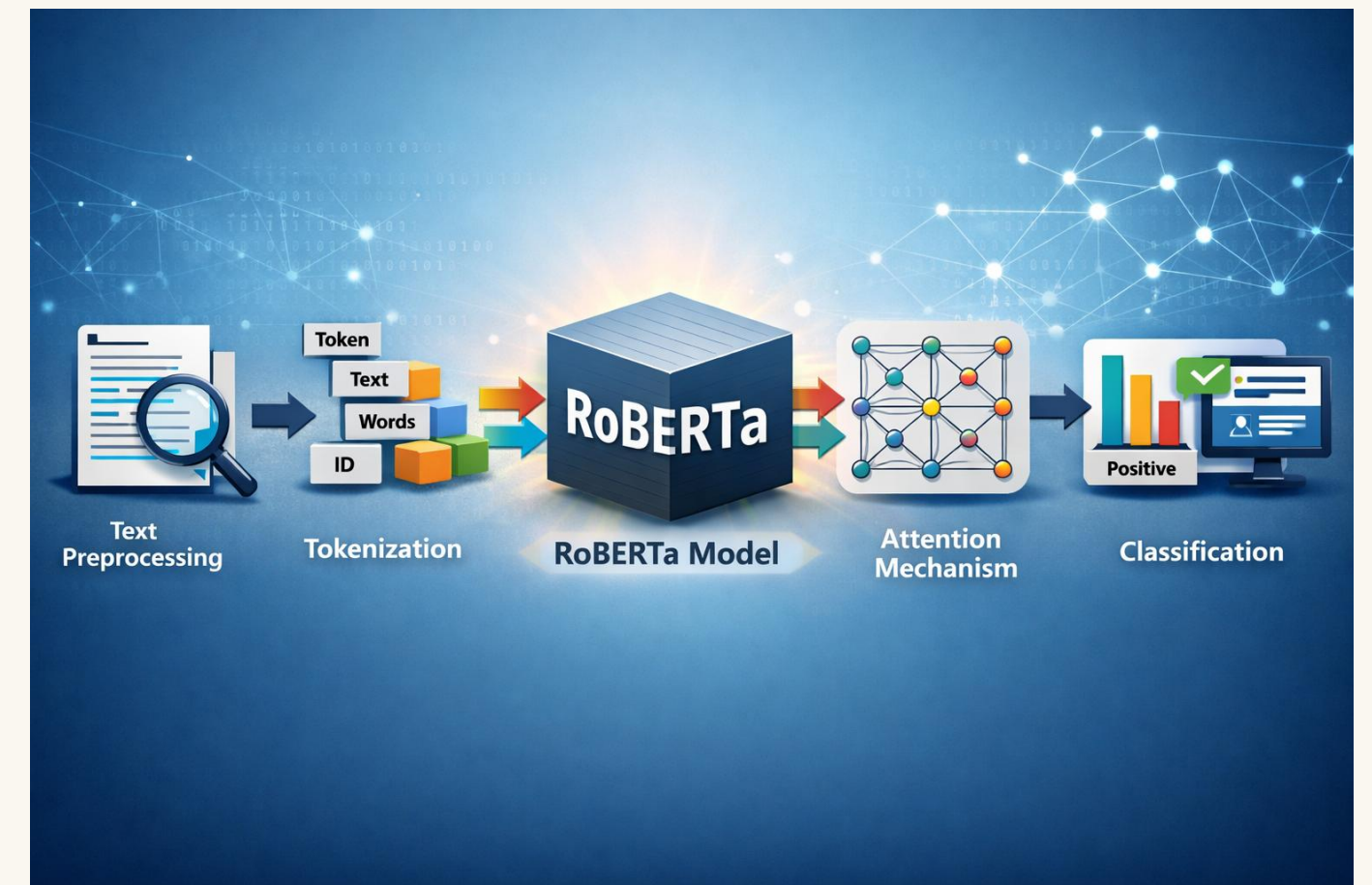
Models:

We compare a zero-shot LLM baseline to a RoBERTa model fine-tuned on our synthetic dataset.

Baseline- GPT-4o-mini (zero-shot, no fine-tuning)

Fine-tuned model:

- Backbone: RoBERTa:
- Training strategies:
 - One-Pass (no class weights)
 - One-Pass (class weights)
 - Pairwise



Approach overview: Pairwise vs One-Pass

Input:

JD X

Skill 1, skills 2,skill 3... skill n

Pairwise

JD X	Skill 1	Implicit/explicit/none
JD X	Skill 2	Implicit/explicit/none
JD X	Skill 3	Implicit/explicit/none
⋮		
JD X	Skill n	Implicit/explicit/none

One-Pass

JD X	Skill 1	Implicit/explicit/none
	Skill 2	Implicit/explicit/none
	Skill 3	Implicit/explicit/none
	⋮	
	Skill n	Implicit/explicit/none

Output:

JD X Skills (non-none)+ implicit/explicit

One-Pass runs once per JD (fast); Pairwise runs once per skill (more accurate, slower).

End-to-End: Data • Models • Metrics

Metrics:

- Precision – Of the skills the model predicted, what fraction are correct.
- Recall – Of the true skills in the resume/JD, what fraction the model successfully finds.
- F1 – A single score that balances Precision and Recall.

Typed scoring: a prediction counts as correct only if both the skill and its label match (IMPLICIT / EXPLICIT).

Example:

GT: {SQL, AWS, Docker, Git}

Pred: {SQL, AWS, Linux, Git, Kubernetes}

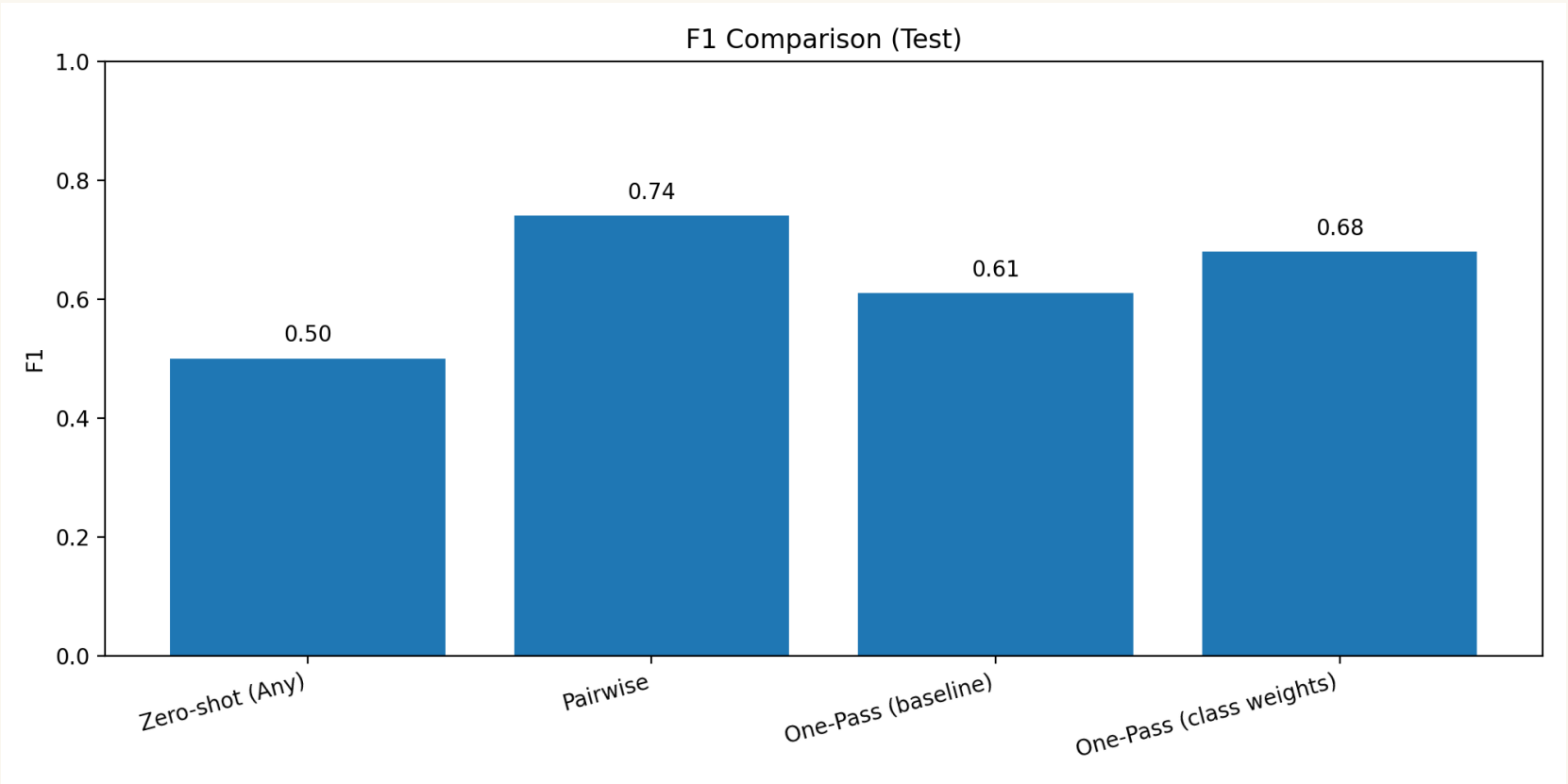
- Precision – $3/5 \rightarrow 60\%$
- Recall – $3/4 \rightarrow 75\%$
- F1 – $2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall}) \rightarrow 0.9/1.35 \approx 0.67 \rightarrow 67\%$

Our Path to High-Quality Results:

- Controlled data: ~3,500 synthetic examples with controlled skill coverage in the text.
- Consistent labeling: each skill in a sample is labeled as EXPLICIT or IMPLICIT.
- Fair evaluation: separate Train / Val / Test splits.
- Systematic comparison: zero-shot baseline vs. RoBERTa under multiple training variants.

Results:

Model	Precision	Recall	F1
GPT-4o-mini (zero-shot)	~55%	~47%	~50%
RoBERTa One-Pass	~66%	~60%	~61%
RoBERTa One-Pass + weights	~65%	~73%	~68%
RoBERTa Pairwise	~67%	~87%	~74%



Conclusion:

Achieved desired results:

- Clear improvement over the zero-shot baseline.
- Pairwise performed best; One-Pass is faster.

What we'd do differently?

- Expand the dataset beyond 3,500 samples to increase coverage and diversity.
- Add additional seeds/runs to verify stability, not rely on a single run.

Future ideas:

- Scale up the skill taxonomy toward a larger global list.
- Evaluate on real resumes/JDs to test generalization.

Thank You